

Hi, I'm Welly Oktariana — Data Analyst

Data Analyst with 2+ year of experience in building machine learning models, performing spatial analysis, and leveraging data visualization techniques to deliver actionable insights that support strategic decision-making and drive business outcomes. Strong interest in data processing and analysis, with hands-on experience working with both large-scale and small datasets across various contexts. Adept at combining spatial and non-spatial data to uncover patterns, identify trends, and translate complex information into clear business recommendations.

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Skills

Technical Skills

Python SQL Streamlit & HTML Machine Learning
GIT Databricks

Analytical Skill

Data Cleaning Data Exploration Data Processing Feature Engineering
Hyperparameter Tuning Spatial Analysis

Projects

House Price Prediction

Objective

Developed a regression-based model to forecast residential property prices by incorporating factors such as location, property size, available amenities, SES data, and demographic data.

Key Outcomes

- provide a model to predict house prices to assist appraisals in the banking industry.

Methods

Python, SQL, Regression Modeling, Feature Engineering, Hyperparameter Tuning

Python LLM SQL OOP

House Price Prediction Using Deep Learning

Objective

Developed a hybrid predictive model for property prices in Surabaya by combining structured transaction data with unstructured satellite images.

Key Outcomes

- Improved predictive accuracy by integrating tabular and spatial features.
- Captured hidden value factors such as vegetation, building textures, and road accessibility.

Methods

Python, TensorFlow, VGG16 (CNN), OpenCV, MinMaxScaler

Python Deep Learning VGG16 Computer Vision

Telco Churn Analysis

Objective

Built a churn prediction model using customer data from a telecommunications company to identify high-risk customers and improve retention strategies.

Key Outcomes

- Detected key behavioral factors driving churn.
- Enabled proactive customer retention through predictive analytics.

Methods

Python, Classification Models, Data Preprocessing, Feature Importance Analysis

Python SQL Scikit-Learn Classification

Recommendation System

Objective

Implemented and compared recommendation algorithms to enhance personalization in user recommendations.

Key Outcomes

- Improved recommendation accuracy using similarity optimization.
- Compared cosine similarity with alternative similarity metrics.

Methods

Python, Collaborative Filtering, Cosine Similarity, Similarity Metrics

Python Recommendation Cosine Similarity SQL

OCR – Document Extraction

Objective

Designed an OCR pipeline to extract structured information from unstructured passport documents.

Key Outcomes

- Automated data extraction from documents, reducing manual input.
- Improved efficiency in handling unstructured text and images.

Methods

Python, OpenCV, Tesseract OCR, Text Parsing, Image Preprocessing

Python OCR Computer Vision NLP

Clustering Flight Dataset with RFM

Objective

Segmented airline customer data using Recency, Frequency, and Monetary (RFM) analysis to identify different customer groups and support marketing strategy.

Key Outcomes

- Discovered distinct customer segments for loyalty program design.
- Enabled data-driven customer retention and targeted campaigns.

Methods

Python, RFM Analysis, K-Means Clustering, Data Standardization

Python Clustering RFM EDA

POI QA Automation Module

Objective

Developed an automated QA module to enhance Point-of-Interest (POI) data accuracy through multi-layer validation, including duplicate detection, location consistency, reviews, common names, and ratings checks

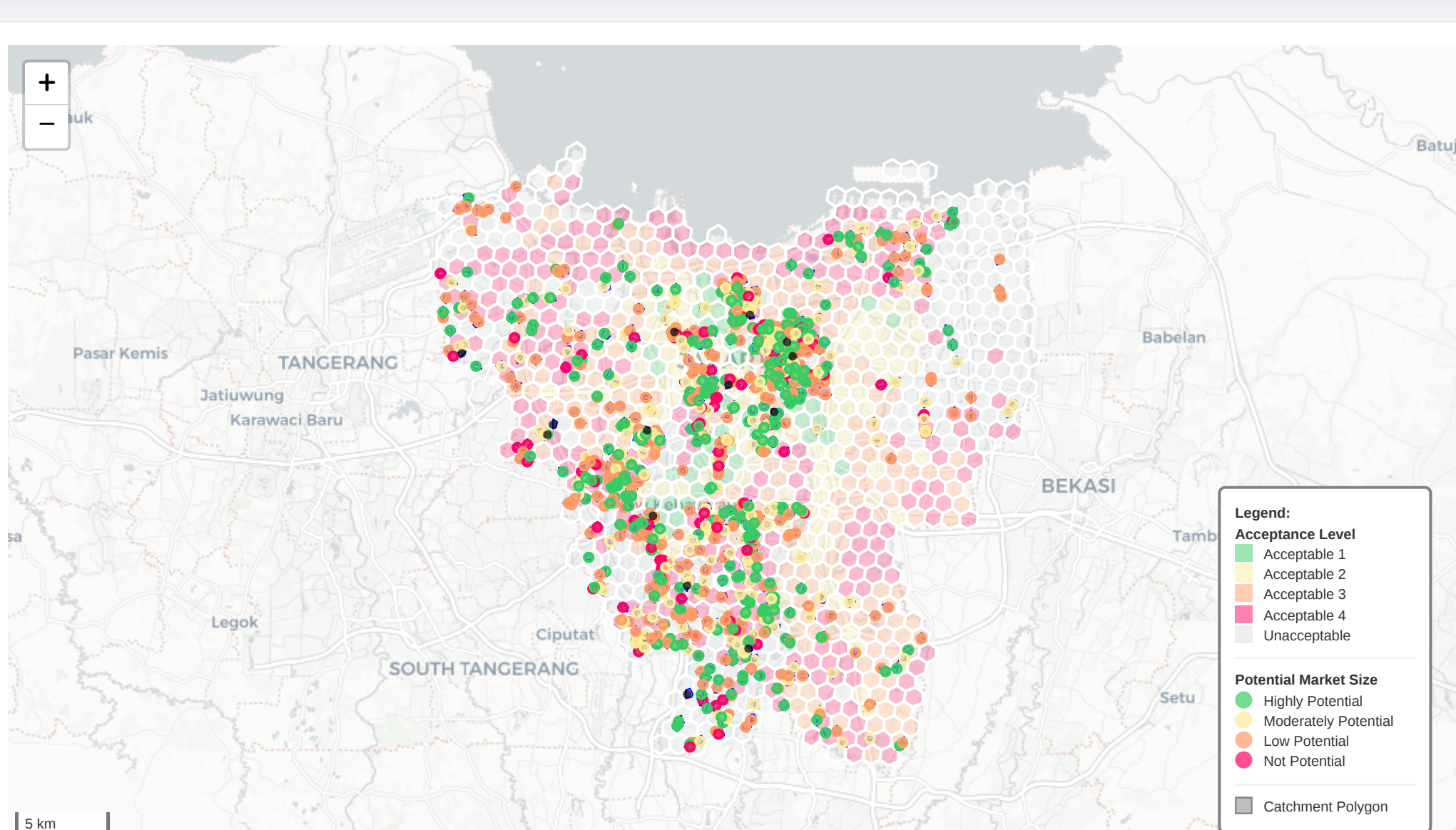
Key Outcomes

- Minimized manual QA effort by automating processing validation tasks.
- Assigned confidence values to POI entries, allowing data to be filtered without reducing the total number of POIs.

Methods

Scoring, Semantic Checks (LLM-based), Text Matching, Geospatial Analysis

Python LLM SQL OOP



Market Potential Analysis

Objective

The project aimed to identify growth opportunities by conducting **Whitespace Analysis** to uncover untapped grocery store potential, performing **Grid Analysis** to evaluate and prioritize attractive areas for strategic placement, and calculating **Market Potential** to estimate store-level opportunities and guide expansion and investment decisions.

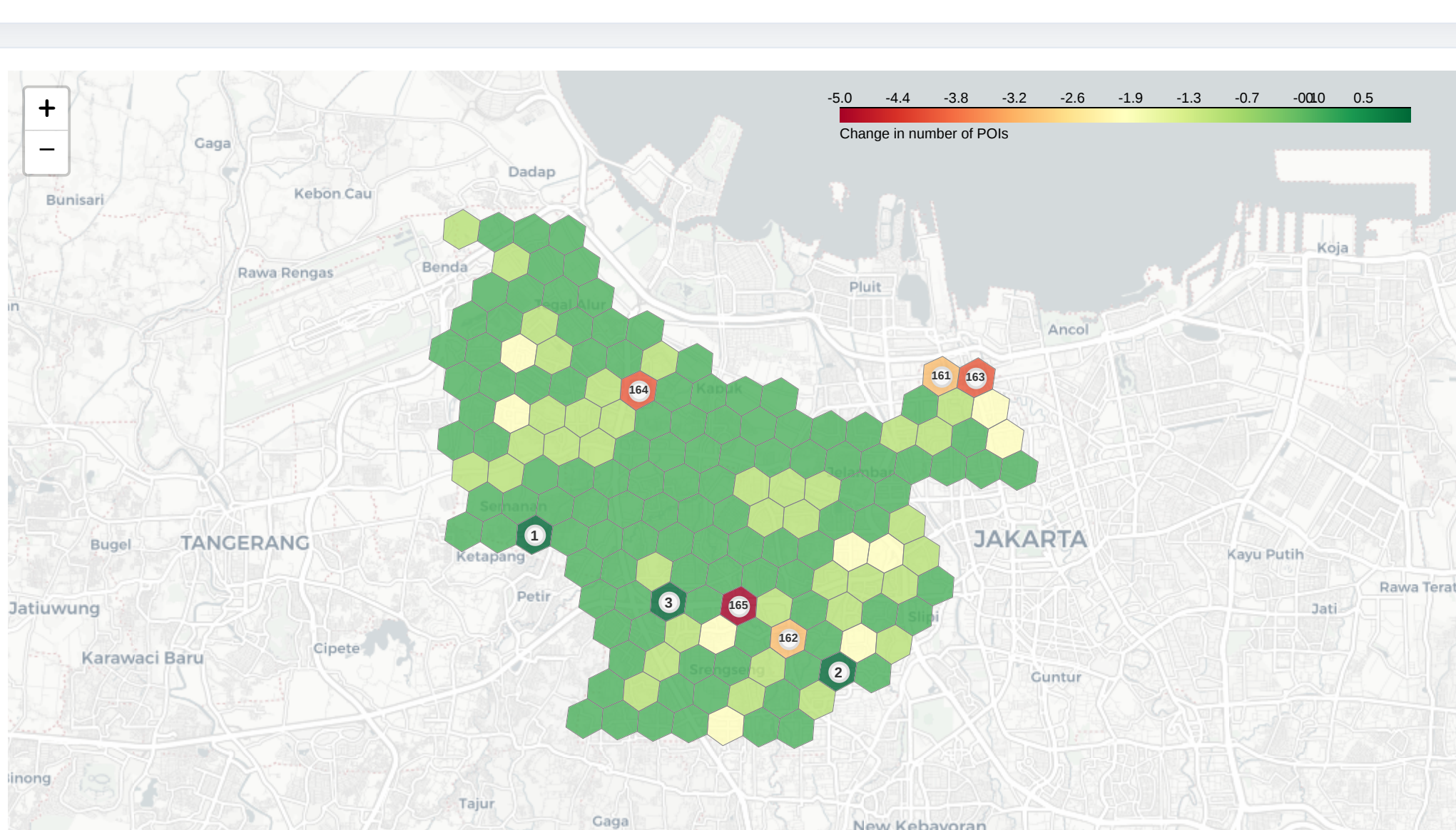
Key Outcomes

- Identified high-potential areas for new store openings, supporting strategic decision-making.
- Highlighted gaps in the existing store network, enabling targeted expansion efforts.
- Provided quantitative estimates of market potential at the store level to inform investment prioritization.

Methods

Whitespace Analysis, Grid Analysis, Market Potential Estimation

Python SQL HTML&CSS Grid Analysis Whitespace Analysis



POI Index Growth Analysis

Objective

Analyzed Point-of-Interest (POI) density changes over time to identify high-growth and declining areas, supporting market expansion and field validation strategies.

Key Outcomes

- Identified micro-level expansion opportunities through high-growth hex clusters.
- Highlighted declining areas for potential reduced activity monitoring.
- Delivered 500 m-level change maps, top/bottom hex tables with percentage shifts, and actionable recommendations for priority zones.
- Created a scalable POI Index framework for continuous monitoring and comparison across periods.

Methods

Spatial Indexes

Python SQL Streamlit Geospatial Analysis

Contact

Let's connect! You can reach me through the following platforms:

✉ wellyoktariana08@gmail.com

🌐 linkedin.com/in/your-profile

🐙 github.com/your-username